CHAPTER ONE

Units and Dimensions

1. Show that $x - x_0 = v_0t + \frac{1}{2}at^2$ is dimensionally correct, where x and x_0 are the final and initial distances, v_0 is the velocity, a is the acceleration and t is the time.

2. A body is falling from rest under gravity. Find the relation between the distance traveled by the body and between the gravitational acceleration and time elapsed.

3. Find the relation between the velocities of transverse waves produced from the vibration of a thin homogeneous string and between the tension in the string and its linear density.

4. Use dimensional analysis to determine how the linear speed of a particle traveling in a circle depends on some or all of the following properties: the radius of the circle, the angular frequency with which the particle orbits about the circle, and the mass of the particle.

5. Use the method of dimensions to find the form of the physical law relating the centripetal acceleration acting on a particle of mass moving on a circle of radius with constant velocity.

6. Newton's law of universal gravitation is represented by $F = G(Mm/r^2)$, where F is the magnitude of the gravitational force exerted by one small object on another, M and m are the masses of the objects, and r is the distance. Find the dimension of the gravitational constant G.

7. Verify the truth of the relation: $dV/dt = Pa^4/(\eta l)$, where dV/dt is the volumetric rate of flow of a liquid of coefficient of viscosity η flowing through a tube of radius a and length l due to pressure difference P.

8. The relationship between kinetic energy K and momentum P is $K = P^2/(2m)$, where m stands for mass. What is the SI unit of momentum?

9. Show which of the following equations are dimensionally correct:

(i) $F_\beta = \rho Vg$, where F_β is the buoyant force acting by a fluid of density ρ on an object of submerged volume V and g is gravitational acceleration.

(ii) $P = P_a + \rho gh$, where P is the pressure at a point placed at a depth h from the surface of a fluid of density ρ, P_a is the atmospheric pressure, and g is gravitational acceleration.

10. Using the method of dimensions, check the validity of: $T = 2\pi\sqrt{(L/g)}$, where T is the period time.